Tamjid Rahat

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Research Interests

Agentic AI and software security; automated program repair (APR); LLM agents for vulnerability detection and remediation; agent evaluation (validators, LLM-as-a-judge); retrieval-augmented generation (RAG) for code; human-in-the-loop learning; formal methods and program analysis.

Education

University of California, Los Angeles (UCLA) 2024

Ph.D. in Computer Engineering • CGPA 4.0/4.0

Dissertation: "Automated Detection and Mitigation of Vulnerabilities in Single Sign-on Implementations."

University of Virginia, Charlottesville, VA 2020

M.S. in Computer Science • CGPA 3.90/4.0

Bangladesh University of Engineering and Technology (BUET) 2016

B.Sc. in Computer Science and Engineering • CGPA 3.84/4.0

Experience

Amazon Web Services (AWS), San Jose, CA 06.2024 – Present

Applied Scientist, AWS Security

- Designed agentic systems for automated program repair (APR) that detect and fix security vulnerabilities across heterogeneous services and codebases at production scale.
- Shipped production code that improved the precision and recall of agent-generated fixes.
- Developed LLM-as-a-judge evaluators to monitor agent output and collect feedback, and used that feedback to improve the agents in the next iteration.
- Built evaluation benchmarks and regression suites to keep fixes consistent across different LLMs and agents.
- Optimized the pipeline for scalability and token efficiency, and used retrieval-augmented generation (RAG) and coding agents such as Claude and Kiro to produce high-quality fixes in production.

United International University, Bangladesh 06.2016 – 07.2018

Lecturer, Computer Science and Engineering

Stochastic Logic Limited, Bangladesh 05.2016 – 07.2018

Quantitative Software Developer

Publications

- **ASE 2025:** T. Al Rahat, Y. Chen, Y. Feng, and Y. Tian. "Automated Repair of OpenID Connect Programs." 40th IEEE/ACM Int. Conference on Automated Software Engineering.
- **ACM CCS 2024:** T. Al Rahat, Y. Feng, and Y. Tian. "AuthSaber: Automated Safety Verification of OpenID Connect Programs." 31st ACM Conference on Computer and Communications Security.
- **ACSAC 2023 [Poster]:** M. Xie, T. Al Rahat, W. Wang, and Y. Tian. "Using Program Knowledge Graph to Uncover Software Vulnerabilities." Annual Computer Security Applications Conference.
- **ACM CCS 2022:** T. Al Rahat, Y. Feng, and Y. Tian. "Cerberus: Query-driven Scalable Vulnerability Detection in OAuth Service Provider Implementations." 29th ACM Conference on Computer and Communications Security.
- **IEEE TPS 2022** [**Best Paper Award**]: T. Saha, T. Al Rahat, N. Aaraj, Y. Tian, and N. Jha. "ML-FEED: Machine Learning Framework for Efficient Exploit Detection." IEEE Int. Conference on Trust, Privacy and Security in Intelligent Systems and Applications.
- **ACM WPES 2022:** T. Al Rahat, M. Long, and Y. Tian. "Is Your Policy Compliant? A Deep Learning-based Empirical Study of Privacy Policies' Compliance with GDPR." 21st Workshop on Privacy in the Electronic Society.
- **VEHITS 2020:** T. Le, I. ElSayed-Aly, W. Jin, S. Ryu, G. Verrier, T. Al Rahat, B. Park, and Y. Tian. "Evaluating the Dedicated Short-range Communication for Connected Vehicles against Network Security Attacks." 6th Int. Conference on Vehicle Technology and Intelligent Transport Systems.
- **IEEE/ACM ASE 2019:** T. Al Rahat, Y. Feng, and Y. Tian. "OAuthLint: An Empirical Study on OAuth Bugs in Android Applications." 34th IEEE/ACM Conference on Automated Software Engineering.
- **ADC 2018:** T. Al Rahat, A. Arman, and M. E. Ali. "Maximizing Reverse k-Nearest Neighbors for Trajectories." 29th Australasian Database Conference.

Technical Skills

- **AI / Agentic:** LLMs and frontier models, LLM agents, retrieval-augmented generation (RAG), LLM-as-a-judge evaluation, agentic workflow orchestration, coding agents (Claude Code, Kiro).
- **Programming Languages:** Python, Java, C/C++, Racket, Intel x86 Assembly, MATLAB.
- **Program Analysis:** WALA, Soot, Ghidra, Pysa, Clang, Z3, Tamarin-prover, Rosette, FlowDroid.
- **Machine Learning:** TensorFlow, Keras, PyTorch, scikit-learn.
- **Web & Data:** JavaScript, PHP, MySQL, SQLite, Oracle.
- **Other:** LaTeX, Bash, HTML, Lisp.

Awards and Grants

- **Google Research Paper Award**, 2023.
- **Qualcomm Innovation Fellowship**, Finalist, 2023.
- **Google Vulnerability Reward**, 2022 ($5,000 for reporting security-critical bugs).
- **Vulnerability Research Grant, Google**, 2022 (security research on Google Identity Services).

- **Graduate Education Fellowship**, UCLA, 2022–2023.
- **UVA Computer Science Fellowship**, University of Virginia, 2018–2019.
- **Bug Bounty**, GoFundMe, 2019.
- **Travel Grant**, Amazon Graduate Research Symposium, 2019.
- **Dean's List Award** and **University Merit Scholarship**, BUET, 2012–2016.

Academic Service

- **Technical Program Committee:** USENIX Security '24 (posters), IEEE S&P '24 (posters), ACM CCS '22 (posters), IEEE S&P '23 (posters).
- **Session Chair:** ACM CCS '22, ACM WPES '22.
- **Peer Reviewer:** CSF '21, ICSTE '22, JSS '22–23.
- **External Peer Reviewer:** S&P '21–24, CCS '21–24, USENIX '21–24, WWW '22, SenSys '21–22.

Media Coverage

- *High-Severity Bug Reported in Google's OAuth Client Library for Java*, The Hacker News, May 2022.